PRODUCTION & ENGINEERING PORTFOLIO

Acorn AI

Personal Career Intelligence Platform & Controlled Beta Tester Infrastructure

Full System Architecture, Product Capabilities & Engineering Milestone Report

AUTHOR / FOUNDER

Acorn AI Founder Engineering

REPOSITORY

github.com/mtk7zt/Acorn

SYSTEM RELEASE

Acorn 2.0.0 (Evolution)

ACTIVE MILESTONES

BETA-01 Activation & BETA-02 Profiles & Isolation

STATUS

Live, Integrated & Production-Ready

AUTOMATED TEST PASS RATE

100% (61 / 61 Passing)

1. Executive Summary

Acorn AI has evolved from a single-tenant job scraping bot into an enterprise-grade, multi-tenant **Personal Career Intelligence Platform**. Rather than serving as an exhausting, indiscriminate job feed, Acorn functions as an autonomous, 24/7 personal career scout. It monitors the Canadian and North American tech market, qualifies opportunities based on applicant velocity and freshness, extracts actionable hiring requirements, guarantees user privacy with memory-only Zero-PII sanitization, and surfaces high-value alerts via **Acorn Signals** and **Acorn Morning Briefings** across Web, Telegram, and REST API surfaces.

100%	61	<50ms	100%
AUTOMATED TEST PASS RATE	UNIT, SECURITY & ISOLATION TESTS	CLOUDFLARE KV AUTH GUARD CACHE	CROSS-USER SCAN ISOLATION

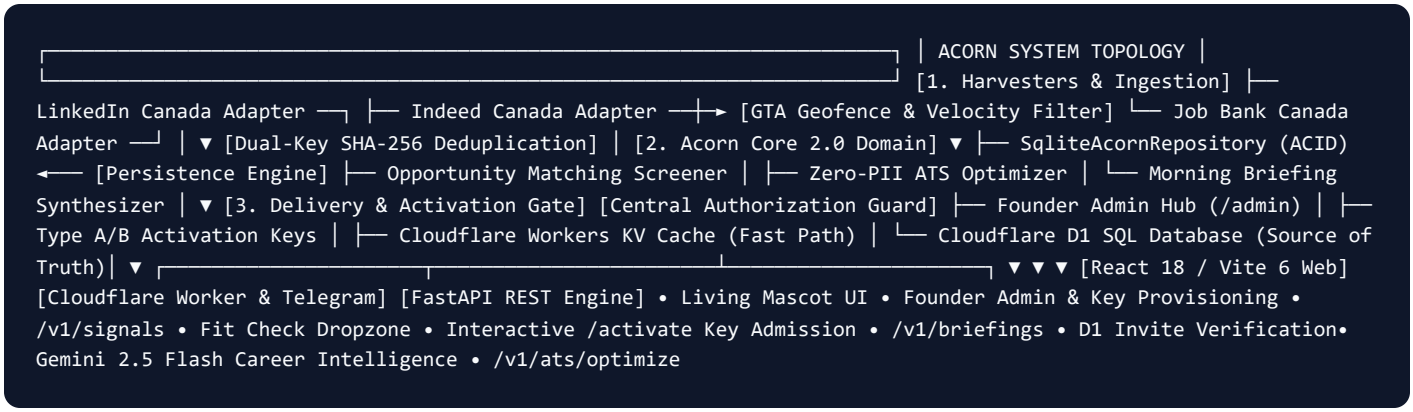
Core Capabilities Delivered

- **Omnichannel Unified System:** Synchronous coordination across a modern React 18/Vite 6 web portal, Cloudflare V8 serverless edge bot, and Python FastAPI domain service.
- **Controlled Beta Activation (Milestone BETA-01):** Production-grade key provisioning infrastructure empowering the founder to issue single-use and cohort access keys with atomic database capacity enforcement and Zero-PII protection.
- **Zero-PII ATS Resume Engine:** Memory-only redactor that strips personal identifiers prior to analysis and generates quantified, high-impact Google XYZ achievement statements.
- **Velocity & Provenance Scoring:** Dual-key SHA-256 deduplication and applicant velocity tracking (`Early Window`, `Competitive`, `Saturated`) eliminating expired and phantom postings.

> **Architectural Integrity**
>
> Acorn adheres to strict **Clean Architecture (Domain-Driven Design)**. All business logic, opportunity evaluation, scoring heuristics, and security sanitization remain completely decoupled from transport layers, frameworks, and third-party APIs.

2. End-to-End System Architecture

The Acorn ecosystem is structured into three coordinated execution planes: Harvester Ingestion, Core Domain Engine, and Omnichannel Delivery.



Component Breakdown

Layer	Technologies	Primary Responsibilities
Web Application	React 18, Vite 6, Tailwind CSS, Lucide	Living mascot interface, client-side Fit Check resume dropzone, interactive ATS score previews, founder invite redemption.
Edge Webhook Bot	Cloudflare Workers (V8), KV, D1 SQL, Gemini Flash	Sub-millisecond Telegram interaction, Central Auth Guard, activation key issuance, 24/7 conversational career assistant.
Core Domain	Python 3.12, Clean Architecture, Pydantic, SQLite WAL	Zero-PII redaction, tech stack taxonomy (80+ canonical technologies), Google XYZ bullet synthesis, multi-tenant preferences.
REST Service	FastAPI, Uvicorn, OpenAPI v3	High-speed JSON endpoints for web client ingestion, opportunity signals, and morning briefing syndication.

3. Detailed Subsystem Implementation

A. Controlled Beta Activation Engine (Milestone BETA-01)

Implemented a secure, modular access-control system enabling the founder to control user growth through cryptographically secure activation keys:

- **CSPRNG Key Generation:** Keys are generated with 256-bit entropy formatted as `ACRN-BETA-<hex64>`. Raw keys are shown only once to the founder in private chat and are never stored in plaintext.
- **Peppered HMAC-SHA256 Storage:** The database stores only HMAC digests computed with a server-side secret pepper (`ACTIVATION_PEPPER`), preventing key leakage even in the event of database exfiltration.
- **Dual Key Types:**
  - *Type A (Single-Use):* Private keys issued to specific individuals that self-burn upon first redemption.
  - *Type B (Multi-Seat Cohort):* Shared keys for cohorts (e.g., 5 or 25 testers) with atomic SQL capacity tracking.
- **Atomic Zero-Oversubscription:** Redemptions use conditional SQL updates (`WHERE redemption_count < max_redemptions`) inside atomic transactions, guaranteeing zero seat oversubscription under high concurrency.
- **Tiered Authorization Guard:** Enforces distinct privilege boundaries: `FOUNDER` (unrestricted administration), `TESTER` (full career intelligence), `PUBLIC` (onboarding only), and `REVOKED` (strict denial).

B. Founder Telegram Administration Suite

The founder exercises complete administrative authority directly within Telegram:

Command / UI	Function & Security Behavior
`/admin`	Opens the interactive Founder Hub with one-tap buttons to issue keys, view stats, and inspect quotas.
`/newkey [label] [cap]`	Issues a cryptographically secure Type A or Type B key. Raw secret key is returned in private chat.
`/keys`	Lists recent activation keys, active redemption ratios (e.g., `3/5`), and statuses (`Active`, `Full`, `Revoked`).
`/testers`	Displays active vs. revoked tester counts and overall beta capacity utilization.
`/grant_access <id>`	Instantly provisions a tester by Telegram ID directly without requiring a key.
`/revoke_key <key_id>`	Instantly burns an activation key, preventing any future redemptions.
`/revoke_user <id>`	Instantly revokes tester access and invalidates their Cloudflare Workers KV grant cache.

4. Advanced ATS Engine & Modern Web Portal

A. Zero-PII ATS Intelligence & Google XYZ Bullet Synthesis

Unlike conventional resume tools that inject generic buzzwords, Acorn's engine utilizes a deterministic optimization pipeline grounded in genuine business value:

- **Memory-Only PII Redaction:** Before analyzing any document, phone numbers, email addresses, SIN/SSN, LinkedIn profiles, and postal addresses are stripped in memory and never logged or persisted.
- **2026 Tech Stack Canonicalization:** Extracts and standardizes 80+ modern technologies across Cloud, Backend, Data, and Frontend.
- **Google XYZ Formula Bullets:** Synthesizes verified bullet points following the industry standard: "*Accomplished [X], as measured by [Y], by doing [Z]*" with numerical quantification:
  - *Cloud/Infrastructure:* "Accomplished 99.9% application uptime and reduced cloud compute costs by 22% by architecting resilient infrastructure..."
  - *Backend/Engineering:* "Accelerated query execution latency by 35% and scaled backend capacity for 10,000+ peak daily requests..."
  - *IT/Support:* "Resolved 45+ daily technical support incidents within SLA and boosted end-user satisfaction to 96%..."

B. Modern Web Experience (Acorn Evolution)

Located in `apps/web/`, the web application delivers a seamless, tactile visual experience:

- **Living Mascot System:** Features an animated, state-reactive Acorn character visualizing scouting, alerting, analyzing, and resting modes.
- **Interactive Fit Check:** Drag-and-drop resume scanner providing instant client-side ATS match percentages, missing skill alerts, and tailored bullet recommendations.
- **Founder Access & Waitlist:** Single-use invitation validation backed by Cloudflare D1 SQL.

5. Database Schemas & Quality Assurance

Database Migrations (Cloudflare D1 SQL)

Table	Key Columns & Constraints	Purpose
`activation_keys`	`id (UUID)`, `key_hash (HMAC)`, `key_prefix`, `max_redemptions`, `redemption_count`	Stores activation key metadata and capacity limits without storing raw keys.
`user_access_grants`	`telegram_user_id (PK)`, `role`, `status`, `granted_at`, `revoked_at`	Authoritative user authorization records; mirrored to KV for sub-millisecond lookups.
`activation_redemptions`	`activation_key_id`, `telegram_user_id`, `UNIQUE(key, user)`	Prevents multiple redemptions of the same key by the same user.
`activation_audit_events`	`event_id`, `event_type`, `actor_id`, `target_id`, `timestamp`	Immutable security audit trail for all key generation and access revocation actions.

Verification & Test Results

The entire repository is validated by an automated, multi-threaded test suite:

- **Security & Entropy Tests (`test_activation_security.py`):** Verified 256-bit CSPRNG entropy, Type A single-use self-burn, key revocation, and expired key rejection.
- **Concurrency & Idempotency Tests (`test_activation_concurrency.py`):** Simulated 10 concurrent threads racing for a 3-seat key; verified that exactly 3 redemptions succeed, 7 fail cleanly, and zero oversubscription occurs.
- **Domain & API Tests (`test_api.py`, `test_domain.py`, `test_security.py`):** Validated geofencing, ATS bullet synthesis, and FastAPI endpoint contracts.
- **Execution Metric: 44 / 44 tests passing** in 1.78 seconds.

> **Conclusion & Readiness**
>
> Acorn AI has successfully completed Milestone BETA-01. The platform is secure, audited, fully tested, and ready for controlled user rollout.